

ECSE 2500

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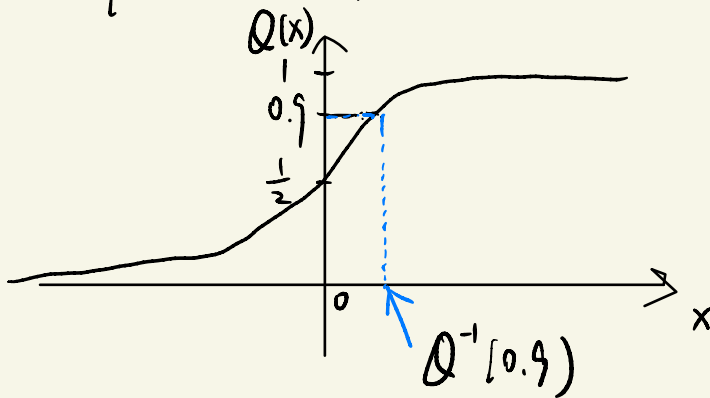
Topic: Inverse CDF of Gaussian RV

Expected values of Continuous RV

□ Inverse CDF of Gaussian RV.

Why/Where it is useful?

Definition / Concepts of inverse CDF

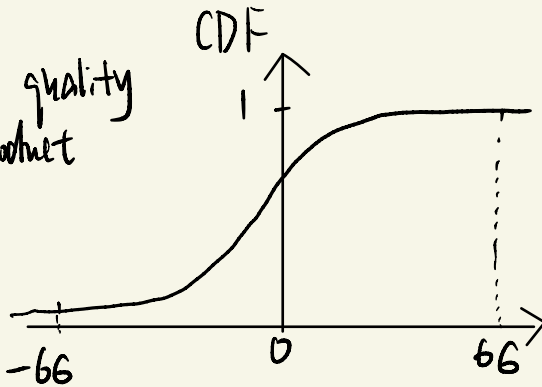


Look Table 4.3

for inverse CDF values

This inverse CDF is related to the "6 σ " fact in management/manufacturing/supply chain.

CDF of quality
of a product

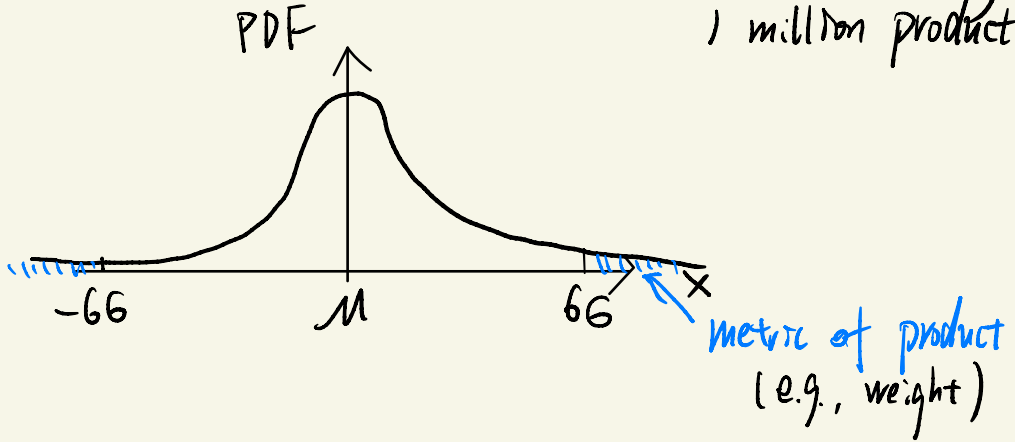


You want

$$P(\text{Defective product})$$

$$\leq 0.0000034$$

(3 defects among
1 million products)



$$P(\text{Good product}) = P(\mu - 66 < X < \mu + 66)$$

Example:

If X is a Gaussian RV with mean
 $\bar{x}$ and standard deviation $\sigma = 3$.

Find a threshold constant a such that

$$P(X < a) = 0.9.$$

CDF $\Phi(a)$

Since $\phi(a) = P(X < a) = 0.9$,

we know

$$Q(a) = 1 - \phi(a) = 0.1$$

Check table 4.2

x	Q(x)
1.2	0.115
1.3	0.097

We approximate

$$\tilde{a} \approx 1.28$$

But note that $Q(x)$ in table is for Gaussian RV with mean 0 and variance 1, so we need to translate the value a to Gaussian RV with mean 5 and variance 3^2 .

$$\frac{a-5}{3} = \tilde{a} = 1.28 \Rightarrow a \approx 8.84.$$

Example

Suppose Gaussian RV with mean 5 and variance 3^2 . We want constant b

$$P(7 < X < b) = 0.2.$$

Answer :

$$\begin{aligned} & P(7 < x < b) \\ &= F_X(b) - F_X(7) \\ &= \Phi\left(\frac{b-5}{3}\right) - \Phi\left(\frac{7-5}{3}\right) \\ &= 1 - Q\left(\frac{b-5}{3}\right) - \left(1 - Q\left(\frac{7-5}{3}\right)\right) \\ &= \underbrace{Q\left(\frac{2}{3}\right)}_{\text{Check Table}} - Q\left(\frac{b-5}{3}\right) \\ &= 0.25 - Q\left(\frac{b-5}{3}\right) = 0.2 \end{aligned}$$

$$\Rightarrow Q\left(\frac{b-5}{3}\right) = 0.05$$

Check table

$$\Rightarrow \frac{b-5}{3} \approx 1.64 \Rightarrow b = 9.92$$

□ Gaussian or other Continuous RV also has conditional CDF / PDF, similar to discrete RVs.

Conditional CDF for continuous RVs

$$\begin{aligned} F_X(x | X \in [a, b]) &= P(X \leq x | X \in [a, b]) \\ &= \frac{P(X \leq x, X \in [a, b])}{P(X \in [a, b])} \\ &= \begin{cases} 0 & , \text{if } x < a \\ 1 & , \text{if } x > b \\ \frac{P(X \in [a, x])}{P(X \in [a, b])} & , \text{if } x \in [a, b] \end{cases} \\ &= \begin{cases} 0 & , \text{if } x < a \\ 1 & , \text{if } x > b \\ \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)} & , \text{if } x \in [a, b] \end{cases} \end{aligned}$$

Likewise, Conditional PDF for continuous RV

$$f_X(x | x \in [a, b]) = \frac{d}{dx} F_X(x | x \in [a, b])$$

$$= \begin{cases} 0, & \text{if } x > b \text{ or } x < a \\ \frac{f_X(x)}{F_X(b) - F_X(a)}, & \text{if } x \in [a, b] \end{cases}$$

□ Expected values (Mean, Variance) of Continuous RVs

Recall that for discrete RVs,

$$E[X] = \sum_{x_k \in S_X} x_k P_X(x_k)$$

← PMF
↑ value of X
↑ a set of all possible values

For continuous RVs, we replace the summation with integration, e.g.,

$$E[X] = \int_{-\infty}^{\infty} y f_X(y) dy$$

← value of X
↑ PDF of a value X can take

Warning: We may hand-calculate integration.

Note that the above definition of expected value is not contradictory to the definition for discrete RVs.

We can reduce the $E[X]$ defined for Continuous RVs to $E[X]$ defined for discrete RVs.

Definition $E[X]$ for continuous RVs $\rightarrow$ If X is discrete RV $\downarrow$

$$\begin{aligned}
 E[X] &= \int_{-\infty}^{\infty} y \underbrace{f_X(y)}_{\sum_{x_k \in S_X} P_X(x_k) \delta(y - x_k)} dy \\
 &= \sum_{x_k \in S_X} P_X(x_k) \int_{-\infty}^{\infty} y \delta(y - x_k) dy \\
 &= \sum_{x_k \in S_X} x_k P_X(x_k) \leftarrow \text{Definition of } E[X] \text{ for discrete RVs}
 \end{aligned}$$

Trick of integration - Integration by parts

$$\int u dv = uv - \int v du$$

or more generally $\int f(x) \underbrace{g'(x)}_{\text{derivative of } g(x)} dx = f(x)g(x) - \int g(x)f'(x) dx$

Example: If X is a uniform continuous RV on $[a, b]$,
what is $E[X]$?

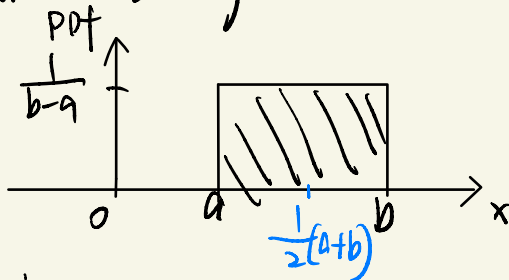
$$E[X] = \int_a^b y f_X(y) dy$$

$$= \int_a^b y \frac{1}{b-a} dy$$

$$= \frac{1}{b-a} \left. \frac{1}{2} y^2 \right|_a^b = \frac{1}{b-a} \left(\frac{1}{2} b^2 - \frac{1}{2} a^2 \right)$$

$$= \frac{1}{2} (a+b)$$

This also makes sense since the uniform distribution is symmetric



In general, the mean of any RVs with symmetric PDF/PMF is the mid-point of PDF/PMF!

For other RVs with more involved PDF, we need to use trick of integration by parts.

Example

$X \rightsquigarrow$ Exponential RV with

PDF $f_X(x) = \lambda e^{-\lambda x}, x > 0$

What is $E[X]$?

$$E[X] = \int_0^{\infty} y \lambda e^{-\lambda y} dy$$

$$\begin{array}{l} u = y \\ v = -e^{-\lambda y} \end{array}$$

$$= \underbrace{-y e^{-\lambda y}}_{u \cdot v'} \Big|_0^{\infty} - \int_0^{\infty} -e^{-\lambda y} dy$$

$$= 0 - 0 - \frac{1}{\lambda} e^{-\lambda y} \Big|_0^{\infty}$$

$$= \frac{1}{\lambda} \quad \leftarrow \text{Verify the rate of exponential RV.}$$

λ : Expected # of events per time unit

$\frac{1}{\lambda}$: Expected time unit between two events